
```
clc;
clear;
close all;

% Parameters
fs = 100000;
t = 0:1/fs:0.02;

fm = 500;           % Message frequency (Hz)
fc = 5000;          % Carrier frequency (Hz)
kf = 100;           % Frequency sensitivity (Hz/V)

% Message Signal
m = cos(2*pi*fm*t);

% Carrier Signal
c = cos(2*pi*fc*t);

% FM Modulation
fm_sig = cos(2*pi*fc*t + ...
    2*pi*kf*cumsum(m)/fs);

% Add Gaussian White Noise
noisy = awgn(fm_sig,10, 'measured'); % 10 dB SNR

% Display Results
figure('Name', 'FM Signal with Gaussian White Noise', ...
    'NumberTitle', 'off');

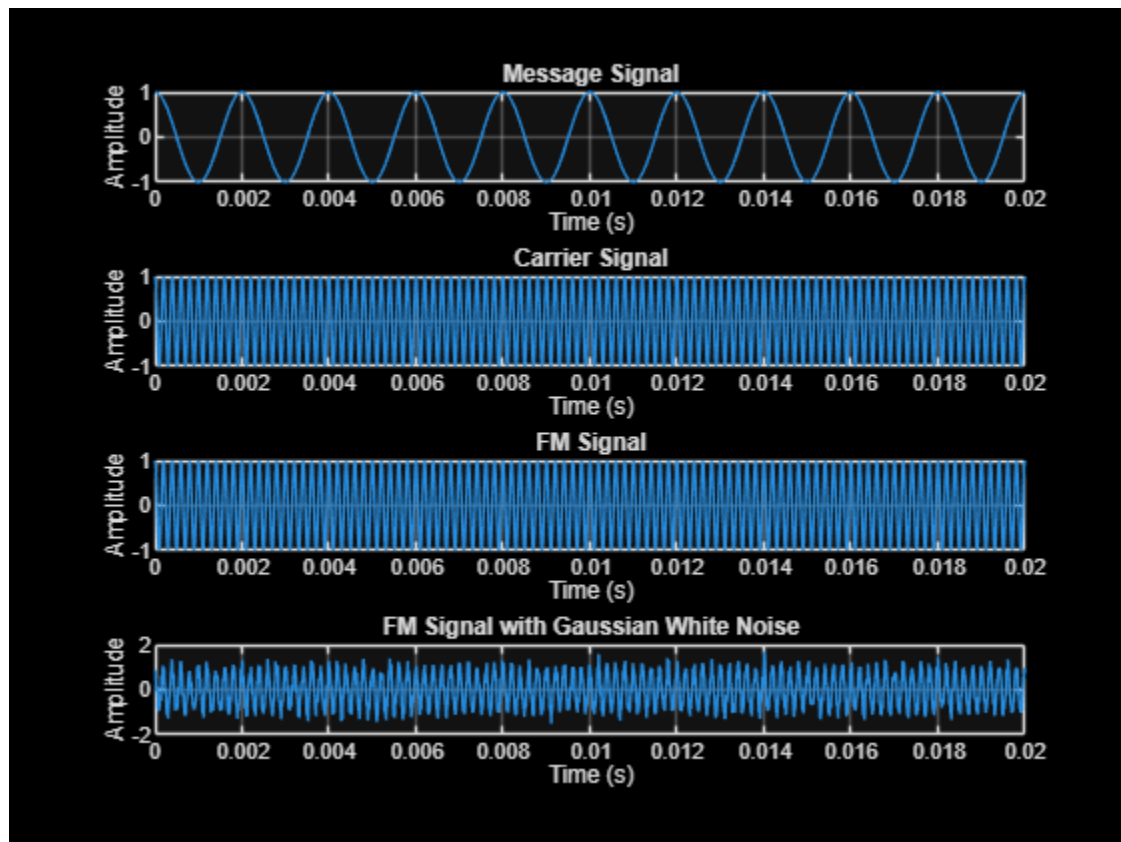
subplot(4,1,1);
plot(t,m);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(4,1,2);
plot(t,c);
title('Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(4,1,3);
plot(t, fm_sig);
title('FM Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(4,1,4);
plot(t, noisy);
```

```
title('FM Signal with Gaussian White Noise');  
xlabel('Time (s)');  
ylabel('Amplitude');  
grid on;
```



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